

Principles of Data Communications and Networks Modeling (CS4865/6865)

Lab Report #2: Introduction to network simulation II

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Exercise 1:

1.1 UDP data flow example

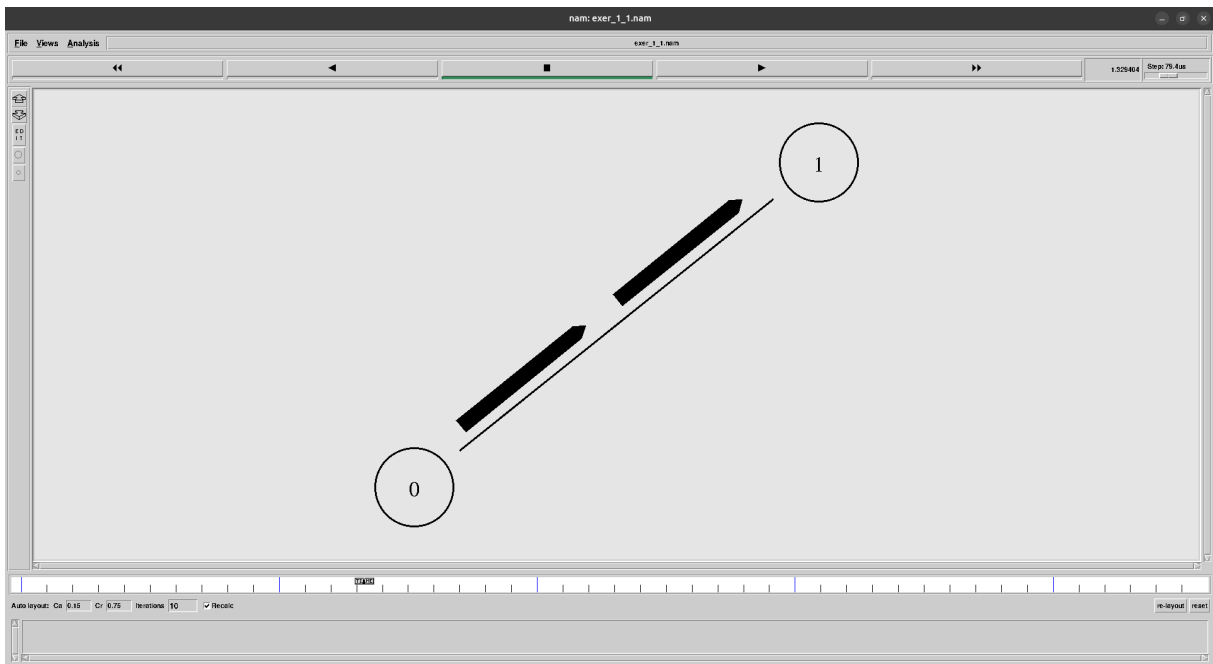


Figure 1: UDP data flow example

1.2 Throughput Calculation

From the trace file, we get:

Timestamp of first packet (send) = 0.5

Timestamp of last packet (received) = 4.51

Total duration of transfer = $(4.51 - 0.5) = 4.01$

Total packets transferred = 801

Size of each packet = 500 bytes

Total data transferred = $801 \times 500 \times 8 = 3,204,000$ bits

Effective throughput = $\frac{3,204,000}{4.1}$ bps $\approx 781,463.41$ bps ≈ 781.46 kbps

Relevant trace file line:

```
1 + -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 1.0
   0 ----- null}
2 r -t 4.513999999999993 -s 0 -d 1 -p cbr -e 500 -c 0 -i 800 -a
   0 -x {0.0 1.0 800 ----- null}
```

1.3 Throughput value comparison

From the *.tcl* file, we get:

Packet size = 500 bytes = 4000 bits

Packet send interval = 0.005 seconds

Theoretical throughput = $\frac{4000}{0.005}$ bps = 800,000 bps = 800 kbps

Comparison:

Theoretical throughput = 800 kbps

Effective throughput = 781.46 kbps

Exercise 2:

2.1 Stop and Wait TCP Flow

The main difference is that in Exercise 2.1 (TCP with window = 1), the traffic behaves like a stop-and-wait protocol which means only one packet is sent at a time, and the sender waits for an acknowledgment before sending the next. In contrast, the UDP/CBR flow in Exercise 1.1 sends packets continuously and quickly, regardless of acknowledgments.

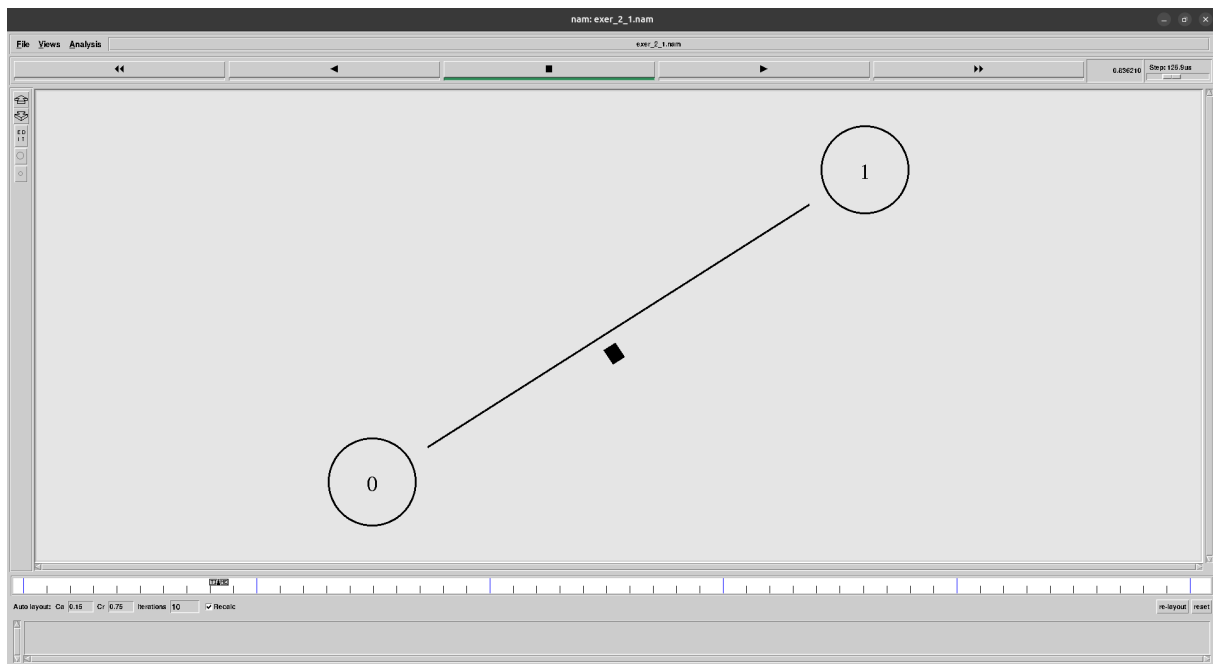


Figure 2: Stop and Wait TCP flow example

Source code:

```

1  # Create a simulator object
2  set ns [new Simulator]
3
4  # Open the nam trace file
5  set nf [open exer_2_1.nam w]
6  $ns namtrace-all $nf
7
8  # Define a 'finish' procedure
9  proc finish {} {
10     global ns nf
11     $ns flush-trace
12     close $nf
13     exec nam exer_2_1.nam &
14     exit 0
15 }
16
17 # Create two nodes
18 set n0 [$ns node]
19 set n1 [$ns node]
20
21 # Create a duplex link between the nodes
22 $ns duplex-link $n0 $n1 1Mb 10ms DropTail
23
24 # Create a TCP agent and attach it to node n0
25 set tcp0 [new Agent/TCP]
26 $tcp0 set packetSize_ 500           ;# Set packet size to 500
27 $tcp0 set window_ 1                 ;# Set max window size to

```

```

1
28 $ns attach-agent $n0 $tcp0
29
30 # Create a TCP Sink agent and attach it to node n1
31 set sink0 [new Agent/TCPSink]
32 $ns attach-agent $n1 $sink0
33
34 # Connect the TCP agent to the sink
35 $ns connect $tcp0 $sink0
36
37 # Create an FTP application and attach it to the TCP agent
38 set ftp0 [new Application/FTP]
39 $ftp0 attach-agent $tcp0
40
41 # Start the FTP traffic at 0.5s
42 $ns at 0.5 "$ftp0 start"
43
44 # Call the finish procedure at 5.0s
45 $ns at 5.0 "finish"
46
47 # Run the simulation
48 $ns run

```

2.2 Throughput Calculation

Since TCP connection has network overhead, I have only considered the actual payload or data that was transferred.

From the trace file, we get:

Timestamp of first packet (send) = 0.52

Timestamp of last packet (received) = 4.99

Total duration of transfer = $(4.99 - 0.52) = 4.47$

Total packets transferred = 182

Size of each packet = 500 bytes

Total data transferred = $182 \times 500 \times 8 = 728,000$ bits

Effective throughput = $\frac{728,000}{4.47}$ bps $\approx 162,863.53$ bps ≈ 162.86 kbps

Relevant trace file line:

```

1 + -t 0.52064 -s 0 -d 1 -p tcp -e 540 -c 0 -i 2 -a 0 -x {0.0
   1.0 1 ----- null}
2 r -t 4.994800000000002 -s 0 -d 1 -p tcp -e 540 -c 0 -i 364 -a
   0 -x {0.0 1.0 182 ----- null}

```

2.3 Throughput value comparison

From the *.tcl* file, we get:

For Link (n0-n1):

Bandwidth (Data Rate): 1Mb = 1 Mbps

Delay = 10ms (one-way)

ForTCP Agent:

Packet Size: 500 bytes

Window Size: 1 packet

Throughput Calculation:

RTT = $2 \times 10 \text{ ms} = 20 \text{ ms}$

Theoretical throughput = $\frac{4000 \times 1}{0.02} \text{ bps} = 200,000 \text{ bps} = 200 \text{ kbps}$

Comparison:

Theoretical throughput = 200 kbps

Effective throughput = 162.86 kbps

Exercise 3:

3.1 Stop and Wait TCP Flow

By increasing the TCP window size from 1 to 2, the sender can transmit two packets before waiting for acknowledgments, instead of just one. This improves throughput and link utilization by reducing idle time between packet transmissions.

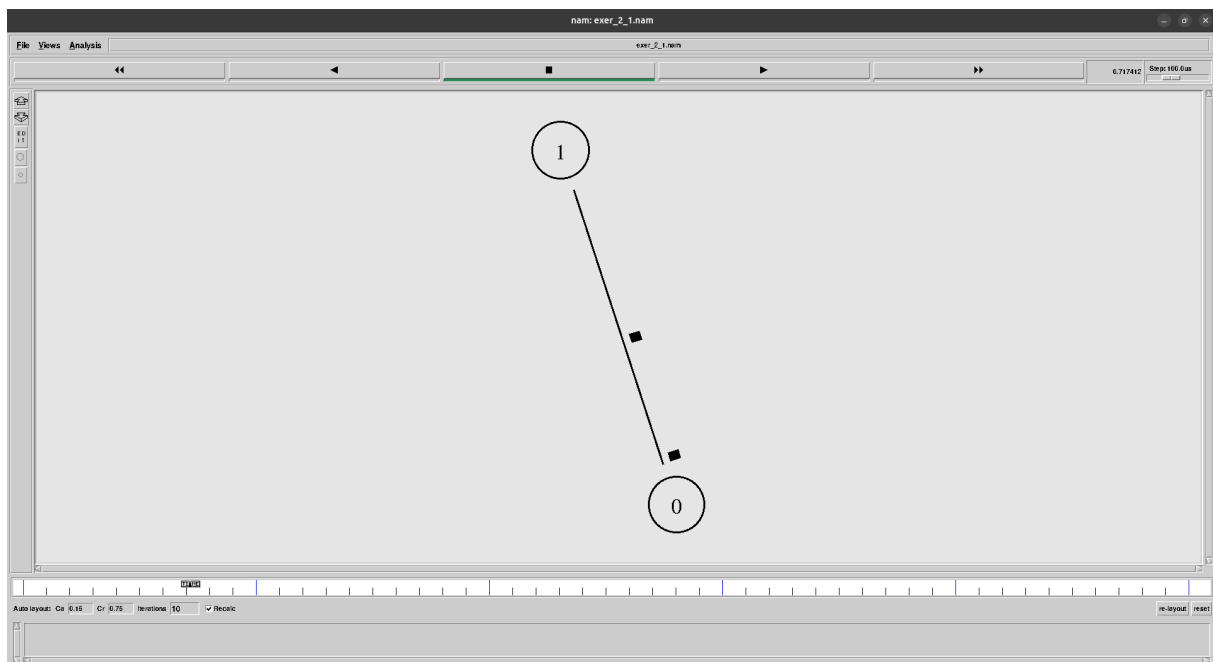


Figure 3: Stop and Wait TCP flow example

Source code:

```
1 # Create a simulator object
2 set ns [new Simulator]
3
4 # Open the nam trace file
```

```

5  set nf [open exer_3_1.nam w]
6  $ns namtrace-all $nf
7
8  # Define a 'finish' procedure
9  proc finish {} {
10     global ns nf
11     $ns flush-trace
12     close $nf
13     exec nam exer_3_1.nam &
14     exit 0
15 }
16
17 # Create two nodes
18 set n0 [$ns node]
19 set n1 [$ns node]
20
21 # Create a duplex link between the nodes
22 $ns duplex-link $n0 $n1 1Mb 10ms DropTail
23
24 # Create a TCP agent and attach it to node n0
25 set tcp0 [new Agent/TCP]
26 $tcp0 set packetSize_ 500           ;# Set packet size to 500
27 $tcp0 set window_ 2                 ;# Set max window size to
    2
28 $ns attach-agent $n0 $tcp0
29
30 # Create a TCP Sink agent and attach it to node n1
31 set sink0 [new Agent/TCPSink]
32 $ns attach-agent $n1 $sink0
33
34 # Connect the TCP agent to the sink
35 $ns connect $tcp0 $sink0
36
37 # Create an FTP application and attach it to the TCP agent
38 set ftp0 [new Application/FTP]
39 $ftp0 attach-agent $tcp0
40
41 # Start the FTP traffic at 0.5s
42 $ns at 0.5 "$ftp0 start"
43
44 # Call the finish procedure at 5.0s
45 $ns at 5.0 "finish"
46
47 # Run the simulation
48 $ns run

```

3.2 Throughput Calculation

Since TCP connection has network overhead, I have only considered the actual payload or data that was transferred.

From the trace file, we get:

Timestamp of first packet (send) = 0.52

Timestamp of last packet (received) = 4.99

Total duration of transfer = $(4.99 - 0.52) = 4.47$

Total packets transferred = 364

Size of each packet = 500 bytes

Total data transferred = $364 \times 500 \times 8 = 1,456,000$ bits

Effective throughput = $\frac{1,456,000}{4.47}$ bps $\approx 325,727.07$ bps ≈ 325.73 kbps

Relevant trace file line:

```
1 + -t 0.52064 -s 0 -d 1 -p tcp -e 540 -c 0 -i 2 -a 0 -x {0.0
   1.0 1 ----- null}
2 r -t 4.999120000000002 -s 0 -d 1 -p tcp -e 540 -c 0 -i 727 -a
   0 -x {0.0 1.0 364 ----- null}
```

3.3 Throughput value comparison

From the *.tcl* file, we get:

For Link (n0-n1):

Bandwidth (Data Rate): 1Mb = 1 Mbps

Delay = 10ms (one-way)

ForTCP Agent:

Packet Size: 500 bytes

Window Size: 2 packets

Throughput Calculation:

RTT = 2×10 ms = 20 ms

Theoretical throughput = $\frac{4000 \times 2}{0.02}$ bps = 400,000 bps = 400 kbps

Comparison:

Theoretical throughput = 400 kbps

Effective throughput = 325.73 kbps

Exercise 4:

4.1 Calculations for the interval parameter

For CBR traffic over UDP, the throughput is determined by:

$$\text{Throughput (bps)} = \frac{\text{Packet Size (bytes)} \times 8}{\text{Interval (seconds)}}$$

For desired interval:

$$\text{Interval} = \frac{\text{Packet Size (bytes)} \times 8}{\text{Desired Throughput}}$$

From the *.tcl* script:

Packet size = 500 bytes

Desired throughput = 200 kbps = 200,000 bps

$$\text{Interval} = \frac{500 \times 8}{200,000} = 0.02 \text{ seconds}$$

4.2 Throughput Calculation

From the trace file, we get:

Timestamp of first packet (send) = 0.5

Timestamp of last packet (received) = 4.51

Total duration of transfer = (4.51 - 0.5) = 4.01

Total packets transferred = 200

Size of each packet = 500 bytes

Total data transferred = $200 \times 500 \times 8 = 800,000$ bits

Effective throughput = $\frac{800,000}{4.01}$ bps $\approx 199,501.25$ bps ≈ 199.50 kbps

Relevant trace file line:

1	+ -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 1.0 0 ----- null}
2	r -t 4.513999999999999 -s 0 -d 1 -p cbr -e 500 -c 0 -i 200 - a 0 -x {0.0 1.0 200 ----- null}

4.3 Achieved Link Utilization

$$\text{Link Utilization (\%)} = \frac{\text{Throughput}}{\text{Link Capacity}} \times 100$$

From the trace file:

Link capacity = 1 Mbps = 1,000 kbps

For exercise 1:

$$\text{Link Utilization (\%)} = \frac{781.46}{1000} \times 100 \approx 7.81\%$$

For exercise 2:

$$\text{Link Utilization (\%)} = \frac{162.86}{1000} \times 100 \approx 16.27\%$$

For exercise 3:

$$\text{Link Utilization (\%)} = \frac{781.46}{1000} \times 100 \approx 32.57\%$$

For exercise 4:

$$\text{Link Utilization (\%)} = \frac{199.5}{1000} \times 100 \approx 19.95\%$$

Comment on link utilizations:

- UDP can send faster, but without congestion control, it may lose packets, especially with DropTail.
- TCP is more reliable, but performance heavily depends on window size and round-trip time.
- Larger TCP window sizes significantly improve link utilization.
- Lower CBR rates result in underutilized links, while higher rates improve utilization but risk packet loss.

References

- [1] Mark Greis, Tutorial for the Network Simulator "ns": <http://www.isi.edu/nsnam/ns/tutorial/index.html>
- [2] TCL description: <http://en.wikipedia.org/wiki/Tcl>
- [3] ns manual: <https://www.isi.edu/nsnam/ns/ns-documentation.html>
- [4] Trace file format: <https://www.isi.edu/websites/nsnam/ns/doc/node289.html>
- [5] nam trace file format: <https://www.isi.edu/websites/nsnam/ns/doc/node616.html>